

**ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)**  
**ORGANISATION OF ISLAMIC COOPERATION (OIC)**  
**Department of Computer Science and Engineering (CSE)**

**QUIZ 1 SET A****DURATION: 30 MINUTES****SUMMER SEMESTER, 2024-2025****FULL MARKS: 20**

**CSE 4205: Digital Logic Design**

**Programmable calculators are not allowed.**

Answer **all 5 (five)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

---

**ID:**

---

1. Simplify the following boolean expression using cut and try method:

4 + 1  
(CO2)  
(PO1)

(i)  $F_1(A, B, C) = \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}\bar{C} + \bar{A}\bar{B}C + A\bar{B}C$

(ii)  $F_2(A, B) = (A \oplus B) + \overline{(A \oplus B)}$

**Solution:**

**(i)**

$$\begin{aligned} F_1 &= \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}\bar{C} + \bar{A}\bar{B}C + A\bar{B}C \\ &= \bar{A}\bar{C}(\bar{B} + B) + A\bar{B}(\bar{C} + C) + \bar{A}\bar{B}C \quad [\text{Distributive}] \\ &= \bar{A}\bar{C} + A\bar{B} + \bar{A}\bar{B}C \quad [\text{Complement, Identity}] \\ &= A\bar{B} + \bar{A}(\bar{C} + \bar{B}C) \quad [\text{Factor}] \\ &= A\bar{B} + \bar{A}(\bar{C} + \bar{B}) \quad [\text{Absorption: } X + X'Y = X + Y] \\ &= A\bar{B} + \bar{A}\bar{C} + \bar{A}\bar{B} \quad [\text{Distributive}] \\ &= \bar{B} + \bar{A}\bar{C} \quad [\text{Absorption}] \end{aligned}$$

**(ii)**

$$F_2 = (A \oplus B) + \overline{(A \oplus B)} = X + X' = 1 \quad [A + A' \text{ form}]$$

**Final Answer:**

$$F_1 = \bar{B} + \bar{A}\bar{C}, \quad F_2 = 1$$

**Rubrics:**

- 2 points for correct grouping/simplification steps for  $F_1$ .
- 2 points for the final simplified form of  $F_1$ .
- 1 point for identifying  $F_2 = 1$ .

2. Perform the following subtraction using the Radix complement (r's complement) method without converting the numbers to another base:

4  
(CO1)  
(PO1)

$$(725)_9 - (368)_9$$

**Solution:**

**Given:**  $(0725)_9 - (0368)_9$

**9's Complement Method (Radix Complement)**

$$8\text{'s complement of } (0368)_9 = (8520)_9$$

$$9\text{'s complement of } (0368)_9 = (8520)_9 + (1)_9 = (8521)_9$$

$$\begin{array}{r} 0 \ 7 \ 2 \ 5 \\ + \ 8 \ 5 \ 2 \ 1 \\ \hline 1 \ 0 \ 3 \ 4 \ 6 \end{array}$$

**Observation:** End carry occurs  $\Rightarrow$  result is positive. Discard carry.

$$\Rightarrow (0346)_9$$

**Rubrics:**

- 1 point for correct 9's complement of the subtrahend.
- 2 points for correct base-9 addition and end carry handling.
- 1 point for the final result.

3. Perform the following addition using BCD addition (you have to convert the numbers to BCD, perform the addition, and convert the result back to decimal):

5  
(CO1)  
(PO1)

$$(567)_{10} + (654)_{10}$$

**Solution:**

$$567_{10} = 0101 \ 0110 \ 0111_{BCD}$$

$$654_{10} = 0110 \ 0101 \ 0100_{BCD}$$

$$\begin{array}{r} 0 \ 1 \ 0 \ 1 \ 0 \ 1 \ 1 \ 0 \ 0 \ 1 \ 1 \ 1 \\ + \ 0 \ 1 \ 1 \ 0 \ 0 \ 1 \ 0 \ 1 \ 0 \ 1 \ 0 \ 0 \\ \hline 1 \ 0 \ 1 \ 1 \ 1 \ 1 \ 1 \ 1 \ 1 \ 0 \ 1 \ 1 \\ + \ 0 \ 1 \ 1 \ 0 \ 0 \ 1 \ 1 \ 0 \ 0 \ 1 \ 1 \ 0 \\ \hline 1 \ 0 \ 0 \ 1 \ 0 \ 0 \ 0 \ 1 \ 0 \ 0 \ 0 \ 1 \end{array}$$

**Final Answer:**

$$1 \ 0010 \ 0010 \ 0001_{BCD} = 1221_{10}$$

**Rubrics:**

- 2 points for correct BCD conversion.
- 2 points for correct BCD addition with corrections.
- 1 point for the final decimal result.

4. A number  $(964)_b$  in an unknown base  $b$  is equivalent to the hexadecimal number  $(A7D)_{16}$ . Determine the base  $b$ .

3  
(CO1)  
(PO1)

#### Solution:

Given:

$$(964)_b = (A7D)_{16}$$

#### Step 1: Convert hexadecimal to decimal

$$(A7D)_{16} = 10 \cdot 16^2 + 7 \cdot 16^1 + 13 \cdot 16^0 = 2560 + 112 + 13 = 2685$$

#### Step 2: Convert $(964)_b$ to decimal and solve

$$9b^2 + 6b + 4 = 2685 \Rightarrow 9b^2 + 6b - 2681 = 0$$

Using the quadratic formula:

$$b = \frac{-6 \pm \sqrt{6^2 - 4 \cdot 9 \cdot (-2681)}}{2 \cdot 9} = \frac{-6 \pm \sqrt{36 + 96516}}{18} = \frac{-6 \pm \sqrt{96552}}{18}$$

Evaluating both roots:

$$b = \frac{-6 + \sqrt{96552}}{18} \approx 16.93$$

$$b = \frac{-6 - \sqrt{96552}}{18} \quad (\text{Not valid})$$

#### Final Answer:

$$b = 16.93$$

#### Rubrics:

- 1 point for correctly converting the two numbers to a known base.
- 1 point for correctly expressing the given numbers into a valid equation.
- 1 point for correctly determining the root of the equation and neglecting the invalid root.

5. Pick the correct option for the following question:

1 × 3  
(CO1,2)  
(PO1)

- (A) Can  $(101101011)_{GRAY}$  and  $(1011011011)_{GRAY}$  be two consecutive numbers in the Gray code sequence?

☐ Yes

☐ No

- (B) Does  $(101101)_2$  maintain even parity?

☐ Yes

☐ No

- (C) Let  $F(A, B, C) = \sum m(0, 3, 5, 7)$ . Is  $\bar{F} = \sum M(1, 2, 4, 6)$  true?

☐ Yes

☐ No

#### Solution:

#### Answers:

(A) No

(B) Yes

(C) No

**Reasoning:**

(A) Consecutive Gray code words must have the same length and differ by exactly one bit; these two have different lengths.

(B)  $(101101)_2$  has four 1s, so the parity is even.

(C)  $\bar{F}$  is the sum of minterms where  $F = 0$  or the product of maxterms where  $F = 1$ ;  $\sum M(1, 2, 4, 6)$  is not correct.

**Rubrics:**

- 1 point for each correct option.